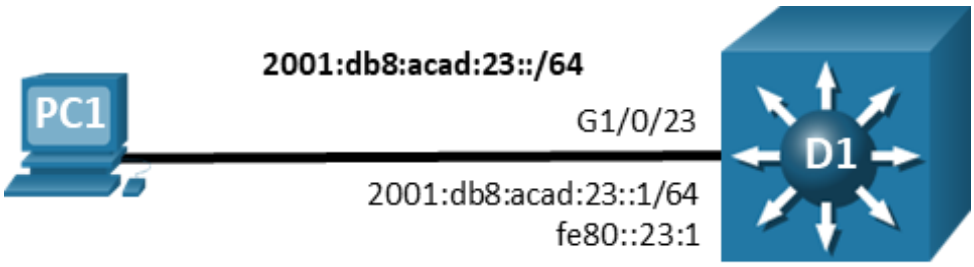
Lab – Implement DHCPv6

Topology



Addressing Table

Device	Interface	IPv6 Global Unicast Address	IPv6 Link-Local Address
R1	GigabitEthernet 1/0/23	2001:db8:acad:23::1/64	fe80::23:1

Objectives

Part 1: Build the Topology and Configure Basic Device Settings and IP Addressing

Part 2: Stateless Address Autoconfiguration (SLAAC)

Part 3: Implement Stateless DHCPv6 Server on a Router

Part 4: Implement Stateful DHCPv6 Server on a Router

Background / Scenario

In this lab, you will explore **IPv6 address configuration mechanisms**, including **SLAAC, stateless DHCPv6, and stateful DHCPv6**, and observe how Router Advertisements influence host behavior. The router **D1** is preconfigured with both an IPv6 **global unicast address (GUA)** and a **link-local address (LLA)** on the LAN interface, but **IPv6 routing is not yet enabled**, so the device is not initially operating as an IPv6 router. While basic IPv4 addressing is present for completeness, **IPv4 connectivity is not required** to complete this lab. You will progressively enable IPv6 routing and DHCPv6 features to examine how hosts dynamically obtain IPv6 addresses, DNS information, and default gateway information under different configuration models.

Note: The switch used with this CCNP hands-on lab is a Cisco Catalyst 9300L using Cisco IOS XE Release 17.12.05 (universalk9 image). Other Layer 3 switches and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the routers and switches have been erased and have no startup configurations. If you are unsure contact your instructor.

Required Resources

- 1 Switch (Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

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Instructions

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on the switch.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram and cable as necessary.

Step 2: Configure basic settings for the switch.

a. Console into the switch, enter global configuration mode, and apply the basic settings and interface addressing using the following startup configuration.

Switch D1

```
hostname D1
!
no ip domain lookup
line con 0
exec-timeout 0 0
logging sync
!
inter g 1/0/23
 no switchport
 description Link to PC
 ip add 10.23.0.1 255.255.0.0
 ipv6 add 2001:db8:acad:23::1/64
 ipv6 add fe80::23:1 link-local
 no shutdown
 exit
!
ip dhcp pool LAN
 network 10.23.0.0 255.255.0.0
 default-router 10.23.0.1
 domain-name example.netacad.com
 exit
!
end
```

b. Save the running configuration to startup-config.

c. PC1 should be able to ping the IPv4 address of the g1/0/23 interface on D1.

Note: If you receive an APIPA address, perform further troubleshooting.

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Part 2: Stateless Address Autoconfiguration (SLAAC)

In this part, you will configure and verify **SLAAC (Stateless Address Autoconfiguration)** so that PC1 can automatically build its own IPv6 address table and learn a default gateway without using DHCPv6. After enabling IPv6 routing on D1, it will send **Router Advertisements (RAs)** that provide the IPv6 prefix for the LAN, allowing the PC to generate a global unicast address and begin IPv6 communication. You will then verify that PC1 learned its IPv6 addressing information and can successfully reach D1 using IPv6.

Step 1: Examine current IPv6 configuration on D1.

At this stage of the lab, **IPv6 routing has not yet been enabled** on D1 (**ipv6 unicast-routing** is not configured), so the router is not sending IPv6 Router Advertisements. The **GigabitEthernet1/0/23** interface is already configured with both IPv4 (not applicable) and IPv6 addressing, including a manually assigned IPv6 global unicast address and link-local address, and will serve as the LAN interface for the SLAAC configuration and verification steps.

```
D1# show run | section interface GigabitEthernet1/0/23
interface GigabitEthernet1/0/23
 description Link to PC
 no switchport
 ip address 10.23.0.1 255.255.0.0
 ipv6 address FE80::23:1 link-local
 ipv6 address 2001:DB8:ACAD:23::1/64
D1#
```

Step 2: Examine IPv6 interface information on G 1/0/23.

The output from **show ipv6 interface** confirms that **GigabitEthernet1/0/23** is operational and correctly configured with both a link-local address and a global unicast IPv6 address. As a result of having an IPv6 link-local address, the multicast address ff02::1 in the following output confirms that the interface belongs to the all-nodes (all-IPv6 devices) multicast group.

At this point, IPv6 is enabled only at the interface level, and because IPv6 routing is not yet enabled on D1, **RAs are not being sent**. The same output will be re-examined after ipv6 unicast-routing is configured to reflect IPv6 routing and Neighbor Discovery behavior associated with SLAAC.

```
D1# show ipv6 interface g 1/0/23
GigabitEthernet1/0/23 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::23:1
  No Virtual link-local address(es):
  Description: Link to PC
  Global unicast address(es):
    2001:DB8:ACAD:23::1, subnet is 2001:DB8:ACAD:23::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
    FF02::1:FF23:1
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds
```

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Step 3: Verify the IPv6 addressing on PC1.

Examine IPv6 addressing on PC1.

```
C:\Users\Student> ipconfig

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : netlab.vm
   Link-local IPv6 Address . . . . . : fe80::6915:c0d:6c6d:2377%14
   Autoconfiguration IPv4 Address. . : 10.23.0.3
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : 10.23.0.1
```

This output shows that the PC successfully received its **IPv4 address via DHCP from D1**, as indicated by the address 10.23.0.3/16 and the default gateway 10.23.0.1.

For IPv6, the PC has automatically generated a **link-local address (LLA)** using a **randomly generated interface identifier which is a host-local operating system behavior**. The %14 at the end of the IPv6 address is the **Windows zone (scope) ID**, which uniquely identifies the local interface on which this link-local address is valid.

At this point, the PC has **no manually configured or dynamically created IPv6 global unicast address (GUA). There is no dynamically created GUA** because PC1 has not yet received any ICMPv6 RAs.

Step 4: Turn on debug and enable IPv6 unicast routing.

a. Before configuring **ipv6 unicast-routing**, turn on **debug ipv6 nd** to observe **Neighbor Discovery (ND) activity** as the router transitions from an IPv6 host to an IPv6 router.

```
D1# debug ipv6 nd
  ICMP Neighbor Discovery events debugging is on
  ICMP ND HA events debugging is ON
```

b. Enable the router as an IPv6 router using the command **ipv6 unicast-routing**.

```
D1# conf t
D1(config)# ipv6 unicast-routing
D1(config)# end
*Jan 21 11:36:03.567: ICMPv6-ND: Node is Router
*Jan 21 11:36:03.567: ICMPv6-ND: (Null0) Periodic RAs Inhibited
*Jan 21 11:36:03.567: ICMPv6-ND: (Null0) Routing change
*Jan 21 11:36:03.567: ICMPv6-ND: (LI-Null0) Routing change
*Jan 21 11:36:03.568: ICMPv6-ND: (GigabitEthernet1/0/23) Routing change
*Jan 21 11:36:03.568: ICMPv6-ND: (GigabitEthernet1/0/23) Periodic RAs Permitted
*Jan 21 11:36:03.568: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) Link RA context
<output omitted>
```

The debug output confirms this change by showing that D1 is now operating as a router and that **periodic RAs**, which were previously inhibited, are now **permitted on GigabitEthernet1/0/23**, allowing SLAAC to begin on the connected PC.

Note: The exact **debug ipv6 nd** output may vary depending on the router model, platform, and IOS/IOS XE version in use.

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Step 5: Examine the router advertisement using debug.

Locate and examine the RA messages in the debug output. To reduce output, disable IPv6 Neighbor Discovery and all other debugging after locating this message.

```
*Jan 21 11:36:04.079: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) send RA to FF02::1
*Jan 21 11:36:04.080: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) Sending RA (1800)
to FF02::1
*Jan 21 11:36:04.080: ICMPv6-ND:   MTU = 1500
*Jan 21 11:36:04.080: ICMPv6-ND:   prefix 2001:DB8:ACAD:23::/64 [LA] 2592000/604800
D1# undebug all
```

(GigabitEthernet1/0/23,FE80::23:1) send RA to FF02::1

- This indicates that D1 is sending an **RA** out **GigabitEthernet1/0/23**.
- The **RA is sourced** (source IPv6 address) from the router's **link-local address FE80::23:1**, which is always used as the source address for RAs.
- The **destination IPv6 address** of the RA is **FF02::1** is the **all-nodes multicast address**, meaning the RA is sent to all IPv6 hosts on the local link.

(GigabitEthernet1/0/23,FE80::23:1) Sending RA (1800) to FF02::1

- This confirms the RA transmission and shows the **Router Lifetime value of 1800 seconds**.
- This lifetime tells hosts how long (in seconds) **D1 should be considered a valid default router**.

MTU = 1500

- The RA includes an MTU option, informing hosts that the link **MTU is 1500 bytes**.
- Hosts will use this MTU for IPv6 communication on this interface, **helping prevent fragmentation issues**.

prefix 2001:DB8:ACAD:23::/64 [LA] 2592000/604800

- This advertises the IPv6 prefix 2001:db8:acad:23::/64 for SLAAC.
- [L] (On-Link flag) - Indicates the prefix is considered directly reachable on the local link.
- [A] (Autonomous flag - Allows hosts to use this prefix to autonomously generate a global unicast address via SLAAC.
- 2592000 = Valid Lifetime (seconds) - How long the address remains valid.
- 604800 = Preferred Lifetime (seconds) - How long the address is preferred before becoming deprecated.

Step 6: Re-Examine IPv6 interface information on G 1/0/23.

After enabling IPv6 unicast routing on D1, re-examine the output of show ipv6 interface g1/0/23 to verify that the interface is now operating as an IPv6 router and actively sending **Router Advertisements (RAs)**.

```
D1# show ipv6 interface g 1/0/23
GigabitEthernet1/0/23 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::23:1
  No Virtual link-local address(es):
  Description: Link to PC
  Global unicast address(es):
    2001:DB8:ACAD:23::1, subnet is 2001:DB8:ACAD:23::/64
  Joined group address(es):
    FF02::1
```

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```
    FF02::2
    FF02::1:FF00:1
    FF02::1:FF23:1
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use stateless autoconfig for addresses.
```

The highlighted fields provide key confirmation of SLAAC behavior:

- **FF02::2** - Indicates that the interface has joined the *all-routers multicast group*, which is required for sending and receiving IPv6 routing-related messages, including sending Router Advertisements.
- **ND router advertisements are sent every 200 seconds** - Shows the interval at which D1 periodically sends RAs on this interface.
- **ND router advertisements live for 1800 seconds** - Specifies the router lifetime advertised in RAs, telling hosts how long D1 should be considered a valid default router. The host refreshes this timer every time it receives an RA.
- **ND advertised default router preference is Medium** - Indicates the **default router preference (DRP)** value advertised to hosts, used when multiple routers are present on the same link. Hosts dynamically populate their **Default Router List** based on the source IPv6 addresses of the RA messages. The DRP (default gateway) can be **high, medium (default), or low.** This helps the host determine which router to use as the default gateway when it receives multiple RA messages.
- **Hosts use stateless autoconfig for addresses** - Confirms that the RA suggests to hosts to use **SLAAC** to automatically generate IPv6 global unicast addresses from the advertised prefix. This confirms that the **[A] (Autonomous flag)** has been set to 1 (default).

Step 7: Re-Examine host IPv6 addressing after SLAAC.

Once again, examine IPv6 addressing on PC1. This output verifies that the PC has successfully **received a Router Advertisement (RA)** and created an **IPv6 global unicast address (GUA)** using the advertised prefix 2001:db8:acad:23::/64.

```
C:\Users\Student> ipconfig

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  : example.netacad.com
   IPv6 Address. . . . . . . . . . : 2001:db8:acad:23:61a4:5032:564f:f329
   Temporary IPv6 Address. . . . . : 2001:db8:acad:23:3cf5:595a:81a4:9ac8
   Link-local IPv6 Address . . . . : fe80::6915:c0d:6c6d:2377%14
   IPv4 Address. . . . . . . . . . : 10.23.0.3
```

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```
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::23:1%14
                                       10.23.0.1
```

Notes:

- The **IPv6 default gateway** is the router's link-local address **fe80::23:1**, and the %14 zone ID matches the link-local address, indicating the same Windows interface is being used.
- The host operating system is using **RFC 4941 (Privacy Extensions for Stateless Address Autoconfiguration)**, generating **randomized interface identifiers** instead of EUI-64 (based on the MAC address), which is evident because it **does not contain ff:fe in the middle of the interface ID**.
- RFC 4941 also creates two addresses: a *stable* **IPv6 address** (sometimes referred to as a public or permanent address), and a **temporary IPv6 address**,
  - This **temporary address** is typically used when the host **initiates outbound communications**, which helps protect user privacy by reducing the ability to track the device over time. To further enhance privacy, the **interface identifier of the temporary IPv6 address changes periodically**, with the timing controlled by the host operating system's privacy extension settings.
  - The **stable IPv6 address** can still be used when **other devices need to initiate connections** to this host. The interface ID can change but is designed to be stable per the prefix.
- When using **SLAAC**, it is normal to see **multiple temporary IPv6 addresses** on a host at the same time. Depending on their state (for example, *preferred* or *deprecated*), older temporary addresses may remain valid for existing connections while newer temporary addresses are created for new outbound traffic, as part of the host's IPv6 privacy extension behavior.

Step 8: Examine DNS information on PC1.

Use the **ipconfig /all** command on PC1, to examine the DNS information for IPv6. Notice that PC1 does not have an IPv6 for valid DNS server.

Note: If you are using Windows, you may see DNS addresses beginning with **fec0:0:0:ffff::.** These are **Windows-generated placeholder DNS addresses** used when no valid IPv6 DNS servers have been learned. These addresses are part of the **deprecated IPv6 site-local address space** (fec0::/10). Site-local addresses were removed from IPv6 standards long ago and **should not be used on today's networks**. Current server and network operating systems do not legitimately advertise these addresses.

Note: "Autoconfiguration Enabled: Yes" indicates that the host operating system is permitted to automatically configure IPv6 addressing. Whether a host actually uses **SLAAC** depends on the **A (Autonomous) flag** in Router Advertisements; if the A flag is set to 0, the host will not create a SLAAC address from that prefix even though autoconfiguration remains enabled.

```
C:\Users\Student> ipconfig /all

Windows IP Configuration

<output omitted>

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : example.netacad.com
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-50-56-A8-68-59
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:23:61a4:5032:564f:f329(Preferred)
```

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```
   Temporary IPv6 Address. . . . . : 2001:db8:acad:23:3cf5:595a:81a4:9ac8(Preferred)
   Link-local IPv6 Address . . . . . : fe80::6915:c0d:6c6d:2377%14(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.23.0.3(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Lease Obtained. . . . . . . . . . : Thursday, January 22, 2026 7:13:35 AM
   Lease Expires . . . . . . . . . . : Friday, January 23, 2026 7:13:34 AM
   Default Gateway . . . . . . . . . : fe80::23:1%14
                                       10.23.0.1
   DHCP Server . . . . . . . . . . . : 10.23.0.1
   DHCPv6 IAID . . . . . . . . . . . : 50334761
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-03-ED-25-00-50-56-A8-68-59
   DNS Servers . . . . . . . . . . . : fec0:0:0:ffff::1%1
                                       fec0:0:0:ffff::2%1
                                       fec0:0:0:ffff::3%1
   NetBIOS over Tcpip. . . . . . . . : Enabled
<output omitted>
```

Step 9: Configure the RA to include an IPv6 DNS server address and verify with debug.

a. Use **debug ipv6 nd** to observe D1 sending the **RA with the IPv6 DNS address.**

```
D1# debug ipv6 nd
  ICMP Neighbor Discovery events debugging is on
  ICMP ND HA events debugging is ON
```

b. Configure the RA to include the IPv6 DNS server address **2001:db8:acad:99::99.** This information is configured on the interface and is specific to the RA message sent out this interface.

```
D1# conf t
D1(config)# interface g 1/0/23
D1(config-if)# ipv6 nd ra dns server 2001:db8:acad:99::9999
```

c. Locate and observe that the RA now includes the DNS server address. The **DNS lifetime** value specifies how long (in seconds) hosts should retain the DNS server information learned from Router Advertisements before it must be refreshed

```
*Jan 22 16:42:59.836: ICMPv6-ND: Allocate ND config on 28
*Jan 22 16:42:59.837: ICMPv6-ND: (GigabitEthernet1/0/23) RA parameter change
*Jan 22 16:43:00.323: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) send RA to FF02::1
*Jan 22 16:43:00.323: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) Sending RA (1800)
to FF02::1
*Jan 22 16:43:00.323: ICMPv6-ND:   MTU = 1500
*Jan 22 16:43:00.323: ICMPv6-ND:   DNS lifetime 600
*Jan 22 16:43:00.323: ICMPv6-ND: DNS Server 2001:DB8:ACAD:99::9999
*Jan 22 16:43:00.324: ICMPv6-ND:   prefix 2001:DB8:ACAD:23::/64 [LA] 2592000/604800
```

d. Disable all debug output.

```
D1(config-if)# end
D1# undebug all
All possible debugging has been turned off
```

Step 10: Re-Examine DNS information on PC1.

Using the **ipconfig /all** command on PC1, re-examine the DNS information for IPv6. Notice that PC1 now has the valid IPv6 address **2001:db8:acad:99::9999** as a **DNS server**.

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```
C:\Users\Student> ipconfig /all

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : example.netacad.com
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-50-56-A8-68-59
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:23:61a4:5032:564f:f329(Preferred)
   Temporary IPv6 Address. . . . . . : 2001:db8:acad:23:3cf5:595a:81a4:9ac8(Preferred)
   Link-local IPv6 Address . . . . . : fe80::6915:c0d:6c6d:2377%14(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.23.0.3(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Lease Obtained. . . . . . . . . . : Thursday, January 22, 2026 7:13:35 AM
   Lease Expires . . . . . . . . . . : Friday, January 23, 2026 7:13:34 AM
   Default Gateway . . . . . . . . . : fe80::23:1%14
                                       10.23.0.1
   DHCP Server . . . . . . . . . . . : 10.23.0.1
   DHCPv6 IAID . . . . . . . . . . . : 50334761
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-03-ED-25-00-50-56-A8-68-59
   DNS Servers . . . . . . . . . . . : 2001:db8:acad:99::9999
   NetBIOS over Tcpip. . . . . . . . : Enabled
<output omitted>
```

Part 3: Implement Stateless DHCPv6 Server on a Router

In this part, you will configure the router to act as a **stateless DHCPv6 server**, which builds on SLAAC rather than replacing it. Hosts will continue to use **SLAAC to generate their IPv6 global unicast addresses and learn the default gateway** from RAs, but DHCPv6 is introduced to provide **additional, generic configuration information**, such as DNS servers and domain names, that is not tied to a specific interface prefix and is often shared across multiple networks. This approach combines the simplicity of SLAAC for addressing with the flexibility of DHCPv6 for distributing common network-wide parameters.

Step 1: Configure D1 as a stateless DHCPv6 server.

a. Remove the IPv6 DNS server information from Router Advertisements on **GigabitEthernet1/0/23**, ensuring that DNS configuration will be provided by **stateless DHCPv6** instead of being advertised directly in the RA.

```
D1(config)# interface g 1/0/23
D1(config-if)# no ipv6 nd ra dns server 2001:db8:acad:99::9999
D1(config-if)# exit
```

b. Create a **stateless DHCPv6 pool** that supplies non-address configuration information, such as **DNS server and domain name**, to IPv6 hosts. A **different DNS server address** is intentionally used so you can clearly verify that the end device is receiving its DNS configuration from the **stateless DHCPv6 server** rather than from RAs.

```
D1(config)# ipv6 dhcp pool IPV6-STATELESS-POOL
D1(config-dhcpv6)# dns-server 2001:db8:acad:88::8888
D1(config-dhcpv6)# domain-name www.example.com
D1(config-dhcpv6)# exit
```

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c. This configuration enables **stateless DHCPv6** on **GigabitEthernet1/0/23** by setting the **O (Other Configuration) flag to 1** in Router Advertisements, instructing hosts to use DHCPv6 to obtain additional configuration information. At the same time, the **A (Autonomous) flag remains set to 1 by default**, allowing hosts to continue using **SLAAC** to generate their IPv6 global unicast addresses while receiving DNS and other network-wide parameters from the stateless DHCPv6 serve

```
D1(config)# interface GigabitEthernet 1/0/23
D1(config-if)# ipv6 nd other-config-flag
D1(config-if)# ipv6 dhcp server IPV6-STATELESS-POOL
D1(config-if)# exit
```

Step 2: Examine the Router Advertisement using debug.

Use **debug ipv6 nd** to display the RAs being sent **after stateless DHCPv6** has been enabled on **GigabitEthernet1/0/23**. This may take **up to 200 seconds** to appear because RAs are sent periodically at that interval unless triggered by a solicitation. Be sure to disable debug after the RA message is displayed.

```
D1# debug ipv6 nd
  ICMP Neighbor Discovery events debugging is on
  ICMP ND HA events debugging is ON
D1#
<output omitted>
*Jan 22 18:13:05.269: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) send RA to FF02::1
*Jan 22 18:13:05.269: ICMPv6-ND:   Other stateful
*Jan 22 18:13:05.269: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) Sending RA
(1800/O) to FF02::1
*Jan 22 18:13:05.270: ICMPv6-ND:   MTU = 1500
*Jan 22 18:13:05.270: ICMPv6-ND:   prefix 2001:DB8:ACAD:23::/64 [LA] 2592000/604800
D1# undebug all
All possible debugging has been turned off
D1#
```

The line **"Other stateful"** indicates that the **O (Other Configuration) flag is set to 1**, suggests to the hosts to use DHCPv6 for additional configuration information.

The "**O**" in the notation **(1800/O)** shows indicates that **O flag** is being advertised as **set to 1**.

The prefix line still displays **[LA]**, confirming that the **A (Autonomous) flag remains set**, allowing hosts to continue using **SLAAC** to generate their IPv6 global unicast addresses even while DHCPv6 is used for other configuration parameters.

Step 3: Examine IPv6 interface information on G 1/0/23.

This output re-examines **GigabitEthernet1/0/23** after configuring stateless DHCPv6, confirming how IPv6 hosts are instructed to obtain their addressing and configuration information.

```
D1# show ipv6 interface g 1/0/23
GigabitEthernet1/0/23 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::23:1
  No Virtual link-local address(es):
  Description: Link to PC
  Global unicast address(es):
    2001:DB8:ACAD:23::1, subnet is 2001:DB8:ACAD:23::/64
  Joined group address(es):
    FF02::1
    FF02::2
```

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```
     FF02::1:2
     FF02::1:FF00:1
     FF02::1:FF23:1
     FF05::1:3
   MTU is 1500 bytes
   ICMP error messages limited to one every 100 milliseconds
   ICMP redirects are enabled
   ICMP unreachables are sent
   ND DAD is enabled, number of DAD attempts: 1
   ND reachable time is 30000 milliseconds (using 30000)
   ND advertised reachable time is 0 (unspecified)
   ND advertised retransmit interval is 0 (unspecified)
   ND router advertisements are sent every 200 seconds
   ND router advertisements live for 1800 seconds
   ND advertised default router preference is Medium
   Hosts use stateless autoconfig for addresses.
   Hosts use DHCP to obtain other configuration.
```

- **FF05::1:3** - This multicast address indicates that the interface has joined the **site-local multicast group for all DHCPv6 servers**, which is used for DHCPv6 communication beyond the local link within the same site.
- **Hosts use stateless autoconfig for addresses** - This indicates that the **A (Autonomous) flag** is set in Router Advertisements, allowing hosts to continue using **SLAAC** to generate their IPv6 global unicast addresses.
- **Hosts use DHCP to obtain other configuration** - This indicates that the **O (Other Configuration) flag** is set, instructing hosts to use **DHCPv6** to obtain additional, non-address configuration (generic) information such as DNS servers and domain names.

Step 4: Verify IPv6 addressing on PC1.

This output verifies the IPv6 configuration on the PC after **stateless DHCPv6** has been enabled, showing that the host continues to use **SLAAC for IPv6 addressing** while receiving **DNS and domain information from the DHCPv6 server**.

```
C:\Users\Student> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-3FR7RKA
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : www.example.com

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : example.netacad.com
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-50-56-A8-68-59
   DHCP Enabled. . . . . . . . . . . : Yes
```

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```
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:23:61a4:5032:564f:f329(Preferred)
   Temporary IPv6 Address. . . . . . : 2001:db8:acad:23:4cce:e25b:425b:4d09(Preferred)
   Link-local IPv6 Address . . . . . : fe80::6915:c0d:6c6d:2377%14(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.23.0.3(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Lease Obtained. . . . . . . . . . : Thursday, January 22, 2026 7:13:35 AM
   Lease Expires . . . . . . . . . . : Friday, January 23, 2026 7:13:34 AM
   Default Gateway . . . . . . . . . : fe80::23:1%14
                                       10.23.0.1
   DHCP Server . . . . . . . . . . . : 10.23.0.1
   DHCPv6 IAID . . . . . . . . . . . : 50334761
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-03-ED-25-00-50-56-A8-68-59
   DNS Servers . . . . . . . . . . . : 2001:db8:acad:88::8888
   NetBIOS over Tcpip. . . . . . . . : Enabled
<output omitted>
```

The following is an explanation of the highlighted output.

- **DNS Suffix Search List** - The appearance of **www.example.com** confirms that the DNS suffix search list is now being learned from the **stateless DHCPv6 server**, which was not available when using SLAAC alone.
- **IPv6 Address (stable GUA** - The stable IPv6 global unicast address **remains unchanged**, confirming that **SLAAC is still responsible for IPv6 address assignment** because the A flag is still set.
- **Temporary IPv6 address**: - temporary IPv6 address is generated using **RFC 4941 privacy extensions** and **may change** if the system is **rebooted, the interface is reset, or when the operating system performs a periodic address rotation** to enhance privacy. If none of these events occur, the temporary address may remain the same for some time, which is why it can appear unchanged during short lab intervals.
- **Link-local IPv6 Address** - The link-local IPv6 address **may change** if the interface is reset or reinitialized. However, it typically remains the same because it is **locally generated by the host** and is **not affected by SLAAC or DHCPv6**. Although modern operating systems use a **non-MAC-based interface identifier**, the link-local address is designed to be stable. On modern operating systems, the link-local IPv6 address typically persists across reboots and only changes if the interface or network configuration is reset.
- **DNS Server**v The DNS server address **2001:db8:acad:88::8888** confirms that the host is now receiving DNS server information from the **stateless DHCPv6 server** rather than from Router Advertisements.

Step 5: Verify DHCPv6 pool on D1.

This output verifies the configuration of the **stateless DHCPv6 pool** on D1 and shows the parameters being advertised to IPv6 hosts.

```
D1# show ipv6 dhcp pool
DHCPv6 pool: IPV6-STATELESS-POOL
  DNS server: 2001:DB8:ACAD:88::8888
  Domain name: www.example.com
  Active clients: 0
```

- **DNS server** - This command displays the IPv6 DNS server address that the stateless DHCPv6 server provides to clients.
- **Domain name** - This command shows the DNS domain name distributed to hosts via stateless DHCPv6.

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- **Active clients: 0** - This command indicates that no DHCPv6 clients currently have an active lease or binding in this pool, which is expected for **stateless DHCPv6** because the router does not track address assignments.

Part 4: Implement stateful DHCPv6 server on a Router

In this part, you will configure **stateful DHCPv6**, which operates in a manner like **DHCP for IPv4** by centrally assigning IPv6 addresses and maintaining state information for each client. One important difference is that **DHCPv6, whether stateless or stateful, does not provide the default gateway**. Instead, IPv6 hosts always learn the default gateway from **RAs**, using the router's **link-local address (LLA)** as the next hop. Depending on the configuration, **SLAAC may still be present** (via the A flag in the RA), but in stateful DHCPv6 the IPv6 address itself is assigned by the DHCPv6 server rather than generated by the host.

Many enterprises choose **stateful DHCPv6** because it allows **centralized control and tracking of IPv6 address assignments**, making it easier to audit which device is using a specific address and to integrate IPv6 addressing with security, logging, and compliance requirements. In these environments, **SLAAC is often disabled,** along with specific security features, so hosts cannot self-generate IPv6 addresses, ensuring that all routable IPv6 addresses are assigned, recorded, and managed by the DHCPv6 server in a manner like traditional IPv4 enterprise networks.

Step 1: Return D1 to a non-DHCPv6 (SLAAC-only) baseline configuration.

This step removes the stateless DHCPv6 configuration from D1 so the interface is returned to a clean IPv6 baseline before configuring **stateful DHCPv6**.

a. Remove the stateless DHCPv6 pool from D1.

```
D1(config)# no ipv6 dhcp pool IPV6-STATELESS-POOL
```

b. Remove stateless DHCPv6 settings from the interface.

```
D1(config)# interface GigabitEthernet 1/0/23
D1(config-if)# no ipv6 nd other-config-flag
D1(config-if)# no ipv6 dhcp server IPV6-STATELESS-POOL
D1(config-if)# exit
```

c. Verify the interface has returned to a non-DHCPv6 configuration.

```
D1# show running-config | section interface GigabitEthernet1/0/23
interface GigabitEthernet1/0/23
 description Link to PC
 no switchport
 ip address 10.23.0.1 255.255.0.0
 ipv6 address FE80::23:1 link-local
 ipv6 address 2001:DB8:ACAD:23::1/64
```

d. You can enter **ipconfig /renew** to reset PC1 ensuring a clean starting point for the stateful DHCPv6 configuration. However, resetting the host is **not required** in most real-world scenarios, as IPv6 hosts will automatically update their configuration when new RAs or DHCPv6 information are received.

Step 2: Configure D1 as a stateful DHCPv6 server.

a. Next, configure a **stateful DHCPv6 pool** that assigns IPv6 addresses directly to clients and maintains state information for each host, similar to DHCP for IPv4. Once again, a **different DNS server address** is intentionally used so you can clearly verify that the end device is receiving its IPv6 address and DNS information from the **stateful DHCPv6 server**, not from SLAAC or a previous configuration.

```
D1(config)# ipv6 dhcp pool IPV6-STATEFUL-POOL
D1(config-dhcpv6)# address prefix 2001:db8:acad:23:deed::/80
```

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```
D1(config-dhcpv6)# dns-server 2001:db8:cafe:77::7777
D1(config-dhcpv6)# domain-name www.example.com
D1(config-dhcpv6)# exit
```

- **address prefix 2001:db8:acad:23:deed::/80** - This command specifies the IPv6 prefix that the DHCPv6 server will use to assign addresses to clients. While a /64 prefix (for example, **2001:db8:acad:23::/64**) is the most common and typical choice in production networks, using a **/80 prefix** here allows the first 16 bits of the interface identifier (**deed**) to be easily recognized. This makes it simple to verify that the IPv6 address was assigned by the stateful DHCPv6 server rather than generated via SLAAC.
- **dns-server 2001:db8:cafe:77::7777** - This command defines the IPv6 DNS server address provided to clients, using a new value to clearly confirm that DNS information is now being supplied by the stateful DHCPv6 server.
- **domain-name www.example.com** - This command s pecifies the DNS domain name that will be distributed to clients via DHCPv6, providing additional network-wide configuration information.

b. This configuration enables **stateful DHCPv6** on **GigabitEthernet1/0/23** by setting the **M (Managed Configuration) flag** in RAs, instructing hosts to obtain their IPv6 addresses from a DHCPv6 server. The DHCPv6 server is then associated with the interface so clients can receive IPv6 addresses and other configuration information from the **IPV6-STATEFUL-POOL**; **by default, the A (Autonomous) flag remains set to 1**, even though address assignment is now managed by DHCPv6 rather than SLAAC.

```
D1(config)# interface GigabitEthernet 1/0/23
D1(config-if)# ipv6 nd managed-config-flag
D1(config-if)# ipv6 dhcp server IPV6-STATEFUL-POOL
D1(config-if)# end
```

Step 3: Examine the Router Advertisement using debug.

This debug output shows **RAs** being sent after enabling **stateful DHCPv6** on the interface, allowing you to verify how hosts are instructed to obtain their IPv6 addressing information. Be sure to disable debug after receiving the RA.

```
D1# debug ipv6 nd
  ICMP Neighbor Discovery events debugging is on
  ICMP ND HA events debugging is ON
D1#
<output omitted>
*Jan 22 21:37:25.246: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) send RA to FF02::1
*Jan 22 21:37:25.246: ICMPv6-ND:   Managed address
*Jan 22 21:37:25.246: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) Sending RA
(1800/M) to FF02::1
*Jan 22 21:37:25.246: ICMPv6-ND:   MTU = 1500
*Jan 22 21:37:25.246: ICMPv6-ND:   prefix 2001:DB8:ACAD:23::/64 [LA] 2592000/604800
D1# undebug all
All possible debugging has been turned off
D1#
```

- **Managed address** - This line indicates that the **M (Managed Configuration) flag** is set in the RA, suggesting to hosts to obtain their IPv6 addresses from a **stateful DHCPv6 server**.
- **The M in (1800/M** - The **M** confirms that the Managed flag is advertised in the RA, signaling stateful DHCPv6 address assignment (the 1800 value represents the router lifetime and can be disregarded here).

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- **The A in [LA** - The **A (Autonomous) flag** remains set, indicating that SLAAC is still permitted by default, even though address assignment is now managed by DHCPv6 (the **L** simply indicates the prefix is on-link and can be disregarded for this discussion).

Step 4: Examine IPv6 interface information on G 1/0/23.

This output verifies the IPv6 behavior of **GigabitEthernet1/0/23** after configuring **stateful DHCPv6**, showing how hosts are instructed to obtain their IPv6 addresses.

```
D1# show ipv6 interface g 1/0/23
GigabitEthernet1/0/23 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::23:1
  No Virtual link-local address(es):
  Description: Link to PC
  Global unicast address(es):
    2001:DB8:ACAD:23::1, subnet is 2001:DB8:ACAD:23::/64
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:2
    FF02::1:FF00:1
    FF02::1:FF23:1
    FF05::1:3
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use DHCP to obtain routable addresses.
```

- **FF05::1:3** - This multicast address indicates that the interface has joined the **site-local multicast group for all DHCPv6 servers and relay agents**, which is used for DHCPv6 communication beyond the local link.
- **Hosts use DHCP to obtain routable addresses** - This line confirms that the **M (Managed Configuration) flag** is set in Router Advertisements, suggesting to hosts to obtain their **global (routable) IPv6 addresses** from the stateful DHCPv6 server rather than using SLAAC.

Step 5: Verifying IPv6 addressing on PC1.

This command displays the IPv6 addressing information currently configured on the PC, allowing you to verify how IPv6 addresses were obtained. The output shows **two stable IPv6 global unicast addresses**: one assigned by **stateful DHCPv6**, identified by the first 16 bits of the interface identifier set to **deed**, and a second address generated using **SLAAC**. Using a **/80 prefix** in the DHCPv6 address pool makes this distinction easy to see, clearly confirming which address was assigned by the stateful DHCPv6 server. **SLAAC** has also generated a **temporary IPv6 address**.

```
C:\Users\Student> ipconfig
```

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```
Windows IP Configuration

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : example.netacad.com
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:23:9f8f:9d75:439:928b
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:23:deed:6a9c:45f:6b4a
   Temporary IPv6 Address. . . . . . : 2001:db8:acad:23:1081:3ff1:9de2:e2c5
   Link-local IPv6 Address . . . . . : fe80::532f:367f:dad2:abc2%14
   IPv4 Address. . . . . . . . . . . : 10.23.0.2
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::23:1%14
                                       10.23.0.1
```

Step 6: Disable SLAAC on G 1/0/23.

This step disables **SLAAC** on **GigabitEthernet1/0/23** so that hosts can no longer generate IPv6 addresses autonomously from RAs. By preventing SLAAC, this configuration ensures that **stateful DHCPv6 is the sole method used for IPv6 address assignment**, making it clear that all IPv6 global unicast addresses are being obtained from the DHCPv6 server rather than being self-generated by the host.

```
D1(config)# interface GigabitEthernet 1/0/23
D1(config-if)# ipv6 nd prefix default no-autoconfig
```

Note: Instead of disabling SLAAC for all advertised prefixes, you could alternatively disable SLAAC for a specific prefix using the following command:

```
D1(config-if)# ipv6 nd prefix 2001:db8:acad:23::/64 no-autoconfig
```

This variation allows more granular control by preventing hosts from using **SLAAC** for a particular prefix while leaving other advertised prefixes unaffected.

Step 7: Re-Examine the Router Advertisement using Debug.

In this step, you use **debug ipv6 nd** to examine the **Router Advertisements (RAs)** sent on **GigabitEthernet1/0/23** and verify that **SLAAC has been disabled**. The debug output confirms that the **A (Autonomous) flag is no longer set**, as indicated by the prefix being advertised with **[L] instead of [LA]**, while the **M (Managed Configuration) flag** remains set, ensuring that hosts must obtain their IPv6 addresses exclusively through **stateful DHCPv6**.

```
D1# debug ipv6 nd
  ICMP Neighbor Discovery events debugging is on
  ICMP ND HA events debugging is ON
D1#
<output omitted>
*Jan 22 22:00:34.438: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) send RA to FF02::1
*Jan 22 22:00:34.438: ICMPv6-ND:   Managed address
*Jan 22 22:00:34.438: ICMPv6-ND: (GigabitEthernet1/0/23,FE80::23:1) Sending RA
(1800/M) to FF02::1
*Jan 22 22:00:34.438: ICMPv6-ND:   MTU = 1500
*Jan 22 22:00:34.438: ICMPv6-ND:   prefix 2001:DB8:ACAD:23::/64 [L] 2592000/604800
D1# undebug all
All possible debugging has been turned off
D1#
```

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Step 8: Re-initialize PC1 and verify IPv6 addressing.

This output verifies that the host is now using **stateful DHCPv6 only** for IPv6 addressing. It has a single IPv6 global unicast address assigned from the DHCPv6 pool (identified by **deed**), no temporary IPv6 addresses are present because **SLAAC is disabled**, and the **default gateway** is still learned via Router Advertisements using the router's **link-local address**.

```
C:\Users\Student> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : example.netacad.com
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:23:deed:6a9c:45f:6b4a
   Link-local IPv6 Address . . . . . : fe80::532f:367f:dad2:abc2%14
   IPv4 Address. . . . . . . . . . . : 10.23.0.2
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::23:1%14
                                       10.23.0.
```

Step 9: Check the IPv6 addressing on PC1.

This output confirms that the host is now receiving **DNS server information and the DNS domain name from the stateful DHCPv6 server**, as shown by the DNS server address (2001:db8:cafe:77::7777) and the DNS suffix search list (www.example.com). Because **SLAAC has been disabled**, this information could only have been provided by the **stateful DHCPv6 configuration**, verifying that the DHCPv6 server is functioning as intended.

```
C:\Users\Student> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-3FR7RKA
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : www.example.com

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : example.netacad.com
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-50-56-A8-68-59
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:23:deed:6a9c:45f:6b4a (Preferred)
   Lease Obtained. . . . . . . . . . : Thursday, January 22, 2026 1:16:19 PM
   Lease Expires . . . . . . . . . . : Saturday, January 24, 2026 12:31:37 PM
   Link-local IPv6 Address . . . . . : fe80::532f:367f:dad2:abc2%14 (Preferred)
   IPv4 Address. . . . . . . . . . . : 10.23.0.2 (Preferred)
```

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```
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Lease Obtained. . . . . . . . . . : Thursday, January 22, 2026 1:16:18 PM
   Lease Expires . . . . . . . . . . : Friday, January 23, 2026 1:16:18 PM
   Default Gateway . . . . . . . . . : fe80::23:1%14
                                       10.23.0.1
   DHCP Server . . . . . . . . . . . : 10.23.0.1
   DHCPv6 IAID . . . . . . . . . . . : 50334761
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-03-ED-25-00-50-56-A8-68-59
   DNS Servers . . . . . . . . . . . : 2001:db8:cafe:77::7777
   NetBIOS over Tcpip. . . . . . . . : Enabled

   Connection-specific DNS Suffix Search List :
                                       www.example.com
<output omitted>
```

Step 10: Verify stateful DHCPv6 activity on D1.

a. This command displays the current status of the **stateful DHCPv6 pool**, allowing you to verify the configured address prefix, DNS information, and client usage. The output confirms that the **IPV6-STATEFUL-POOL** is using the specified **/80 address prefix**, shows that **one IPv6 address is currently allocated and in use**, and indicates **one active DHCPv6 client**, verifying that stateful DHCPv6 address assignment is operating correctly.

```
D1# show ipv6 dhcp pool
DHCPv6 pool: IPV6-STATEFUL-POOL
  Address allocation prefix: 2001:DB8:ACAD:23:DEED::/80 valid 172800 preferred 86400
(1 in use, 0 conflicts)
  DNS server: 2001:DB8:CAFE:77::7777
  Domain name: www.example.com
  Active clients: 1
```

b. This command displays the **stateful DHCPv6 bindings**, allowing you to verify which IPv6 address has been assigned to a specific client and how long that address remains valid. The output shows the **client's link-local address (FE80::532F:367F:DAD2:ABC2)** used for DHCPv6 communication, the **IPv6 address assigned from the DHCPv6 pool (2001:DB8:ACAD:23:DEED:6A9C:45F:6B4A)**, and the associated **preferred and valid lifetimes**, confirming that the address is actively leased to the client and will expire at the indicated time if not renewed.

```
D1# show ipv6 dhcp binding
Client: FE80::532F:367F:DAD2:ABC2
  DUID: 000100013103ED25005056A86859
  Username : unassigned
  VRF : default
  IA NA: IA ID 0x03000C29, T1 0, T2 0
    Address: 2001:DB8:ACAD:23:DEED:6A9C:45F:6B4A
            preferred lifetime 86400, valid lifetime 172800
            expires at Jan 24 2026 09:19 PM (169097 seconds)
```

On **Cisco IOS XE**, the **default stateful DHCPv6 lease timers** are:

- Preferred lifetime - 86,400 seconds (24 hours)
- Valid lifetime - 172,800 seconds (48 hours)

The remaining lifetime of the **DHCPv6 lease** is the **Valid lifetime of 48 hours** expressed in two ways:

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- **expires at Jan 24 2026 09:19 PM** is the wall-clock time when the **valid lifetime** will end if the lease is not renewed
- **(169097 seconds)** is the **remaining number of seconds** until that expiration moment, counted from when the command was issued.

c. This output provides a **summary of DHCPv6 message activity** on D1, confirming that stateful DHCPv6 exchanges are occurring successfully.

- **Messages received / sent** – Shows that DHCPv6 communication is active, with no messages discarded or failed.
- **SOLICIT / ADVERTISE** – Indicates the client discovered the DHCPv6 server and the server responded, starting the address assignment process.
- **REQUEST / REPLY** – Confirms the client requested an IPv6 address and the server successfully assigned it.
- **CONFIRM** – Shows the client verified that its assigned IPv6 address is still valid on the link, which commonly occurs after a restart or interface reset.

Overall, this output confirms **normal and successful stateful DHCPv6 operation** with no errors

```
D1# show ipv6 dhcp statistics
Messages received                4
Messages sent                    4
Messages discarded               0
Messages could not be sent       0

Messages                         Received
SOLICIT                          1
REQUEST                          1
CONFIRM                          2

Messages                         Sent
ADVERTISE                        1
REPLY                            3
```

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